USN | | | | | | | | | | [23MSCH1H04]

FIRST SEMESTER UNIVERSITY EXAMINATION, JANUARY - 2025

ANALYTICAL CHEMISTRY

Time: 03 HOURS **Maximum Marks:50**

		Blooms Level	CO	Marks
	SECTION – A			
1.	**Answer any FOUR of the following questions.**			**4×5=20**
a.	What are determinate and indeterminate errors?	1	1	**05**
b.	Explain the principle and separation mechanism in chromatographic techniques.	2	2	**05**
c.	Compare the thermal analysis methods TGA, DTA and DSC.	2	4	**05**
d.	Explain the competitive and non-competitive assay formats.	2	5	**05**
e.	Why does the peak current in cyclic voltammetry plots increase with an increase in scan rate? Additionally, outline the criteria for selecting solvents and electrolytes for cyclic voltammetry measurements.	2	6	**05**
	SECTION – B			
2.	**Answer any TWO of the following questions.**			**2×9=18**
a.	Write a note on the role of the following; **i.** Pumps with neatly labelled diagram **ii.** Types of injection valves with relevant diagrams in HPLC.	2	3	**09**
b.	Calculate the pH during the titration of 49.00 mL of 0.100 M NaOH with 0.1000M HCl after the addition of **i.** 0.0 ml **ii.** 49 ml and **iii.** 51 ml of HCl. Draw the titration curve.	5	4	**09**
c.	Discuss the limitations of classical analytical chemistry in the analysis of biomolecules and how they are solved in bio-analytical chemistry.	2	5	**09**
	SECTION – C			
3.	**Answer the following question.**			**1x12=12**
a.	**i.** What is the object of the sampling step in an analysis?	2	1	**04**
	ii. What are rate theory and plate theory of column chromatography? Explain in detail.	5	2	**04**
	iii. Describe the Koutecky-Levich Analysis for an electron transfer reaction with slow kinetics. How can you use a rotating disc electrode (RDE) to determine the diffusion coefficient of an analyte and the standard rate constant for an electrochemical reaction?	2	6	**04**